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Volkswagen's Emissions Scandal:

How Could It Happen?

2nd Prize in the "Corporate Sustainability" track of the oikos Case Writing Competition 2018

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This case was written by Erin McCormick under the supervision of N. Craig Smith, the INSEAD Chaired Professor of Ethics and Social Responsibility. It is intended to be used as a basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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Exhibit 1. A Greenpeace protester outside Volkswagen's headquarters. Source: Associated Press

Volkswagen management called it "Strategy 2018." The automaker first rolled out its plan to achieve world domination of the passenger car market in 2007, with the creation of new models to achieve sales targets, especially in the US, Russia and China. With the support of chairman Ferdinand Piëch, CEO Martin Winterkorn set the sales target at 10 million cars a year – an ambitious increase on the 6 million VW was then selling. The key to making the plan work would be to market "clean diesel" engines to the environmentally-conscious segment of the American public. If it hit its goal, the company would overtake GM and Toyota to become the world's biggest carmaker.

Eight years later, CEO Winterkorn was forced to resign in disgrace, and Oliver Schmidt, VW's former top emissions compliance manager for the United States, appeared before a US judge in handcuffs and orange prison jumpsuit, charged with fraud. The company was embroiled in an environmental scandal in which it was revealed that for nearly a decade it had been programming the computer code in its cars to foil emissions tests. Millions of its cars around the world were polluting as much as 40 times the legal limit.[1]

As of September 2017, seven other mid-level VW executives faced prosecution in the US. German prosecutors were mounting another investigation, homing in on Winterkorn and 35 other company managers.[2] The company's stock had crashed. The automaker had been forced

1 Emissions laws varied from country to country. Emission regulations governing nitrogen oxides in Europe were less strict than the US.

2 https://www.bloomberg.com/news/articles/2017-01-27/prosecutors-extend-vw-criminal-probe-to-former-ceo-winterkorn

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to put aside more than $30 billion to pay for US criminal and civil penalties, reimbursements to US customers, worldwide repair costs, and other charges.[3]

While the company admitted guilt, VW senior management blamed the problem on rogue engineers who had operated without their knowledge. But investigators representing the German, American and even South Korean governments were circling to put the blame on the people at the very top, keen to know who had allowed such a deception to extend so far and for so long. "This is a case of deliberate, massive fraud perpetrated by Volkswagen management," said Judge Sean F. Cox of Federal District Court in Detroit as he imposed a $2.8 billion fine. "We don't know how far up this goes. We hope the Justice Department will find and prosecute those responsible."[4]

Volkswagen

Early History

In the 1930s, Adolf Hitler, the Chancellor of Germany, commissioned Ferdinand Porsche to build a "car for the people" or "Volkswagen."[5] Porsche, an unconventional thinker with a genius for engineering, drafted plans for what would later be known as the Volkswagen Beetle. In 1938, Hitler laid the cornerstone of the Volkswagen factory in Wolfsburg Germany but World War II broke out before the first cars could be delivered and the factory was turned into a manufacturing plant for military vehicles, later sustaining heavy bombing.

After the war, Porsche was imprisoned for working with the Nazis. When freed, he came back to the bombed-out factory and helped get it going again under British management to manufacture the VW Beetle he had designed. Eventually he moved on and created the Porsche company, starting another automotive empire. Ironically, after its fascist start, the Beetle went on to become an icon of the sixties leftist counter culture in the US. With over 21 million produced, the Beetle was the best-selling and longest-running single car design in automotive history and propelled Volkswagen into the ranks of the world's top automakers.

By the early 2000s, Volkswagen was struggling to keep a grip on the US auto market. It sold more cars in Europe than any other brand, in part because its diesel models had found favour with European consumers thanks to government incentives. Since they released fewer greenhouse gases than petrol vehicles, officials made diesel fuel and vehicle registration tax cheaper. In the US, there were no such incentives. As Volkswagen's sales there slumped, Toyota was stepping up the competition with its affordable and pro-environmental Prius and other hybrid models.

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3 Jan Schwartz, Victoria Bryan, "VW's Dieselgate bill hits $30 bln after another charge," Reuters, September 29, 2017, https://www.reuters.com/article/legal-uk-volkswagen-emissions/vws-dieselgate-bill-hits-30-bln-after-another-charge-idUSKCN1C4271

4 Jack Ewing, "VW Engineers Wanted O.K. From the Top for Emissions Fraud, Documents Show," New York Times, May 17, 2017, https://www.nytimes.com/2017/05/17/business/volkswagen-muller-diesel-emissions.html

5 Wall Street Journal, "The History of Volkswagen, 'The People's Car'," September 28, 2015, https://www.youtube.com/watch?v=HhH-oWHzzvQ

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Unusual Corporate Structure

The company was governed by a strange mix of family control, government ownership and labour influence, which made its board particularly ineffectual, according to Professor Markus Roth of Philipps-Universität Marburg, an expert on corporate governance: "It's been a soap opera ever since it started," was his verdict.[6]

The structure of VW's board was so tilted toward labour that it was hard for the company to focus on business priorities, experts said. Under German law, the supervisory board was supposed to work hand in hand with a management board of top company executives to oversee strategic direction. Like all German companies, half the seats on the 20-member supervisory board went to labour representatives, and half to shareholders. Labour's role was further bolstered by the fact that two of the shareholder seats went to government representatives from the state of Lower Saxony, a part owner of VW. They inevitably backed workers, giving labour outsized control.[7]

A former VW executive told the *New York Times* that there was a constant push to save jobs: "There's no other company where the owners and the unions are working so closely together as Volkswagen. What management, the government and the unions all want is full employment, and the more jobs, the better. …That's behind the push to be No. 1 in the world. They'll look the other way about anything."[8]

The board was marked by a lack of independent decision-making: only one board member (an outsider) was neither a shareholder nor an employee representative. Critics said the supervisory board seemed to be there for appearances only – with managers circumventing it to thrash out agreements with the unions before issues ever came to the board. "The board was only really there for show," one former VW executive told the *Financial Times*. "They lacked the ability to ask any deep technical questions – and you see that in the current scandal."[9]

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6 James Stewart, "Problems at Volkswagen start in the Boardroom," New York Times, September 24, 2015, https://www.nytimes.com/2015/09/25/business/international/problems-at-volkswagen-start-in-the-boardroom.html

7 Sarah Gordon "VW's board needs to look in the mirror," May 18, 2016, Financial Times, https://www.ft.com/content/76ba9a3e-1c42-11e6-b286-cddde55ca122

8 James Stewart, "Problems at Volkswagen start in the Boardroom," New York Times, September 24, 2015, https://www.nytimes.com/2015/09/25/business/international/problems-at-volkswagen-start-in-the-boardroom.html

9 Richard Milne, "Volkswagen System Failure," November 4, 2015, Financial Times, https://www.ft.com/content/47f233f0-816b-11e5-a01c-8650859a4767

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A Domineering Management Culture

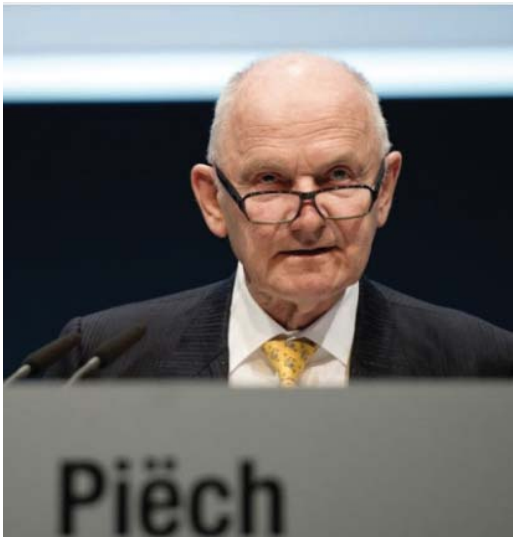

Exhibit 2. Ferdinand Piëch.
Source: Handelsblatt

Perhaps because of its ineffectual board, the company had a series of domineering managers who ran the organization with a rigid, top-down style. Winterkorn (and long before him Ferdinand Piëch, grandson of the Beetle's inventor), were said to have ruled with an iron fist. They set high targets for sales and engineering developments. Anyone who failed to meet the targets or dared to speak against them feared reprisals. Whereas Piëch could upend a manager with a silent scowl, subordinates criticized Winterkorn for being a yeller. "If you presented bad news, those were the moments that it could become quite unpleasant and loud and quite demeaning," said a former VW executive.[10] "VW had this special culture," said Arndt Ellinghorst, a former Volkswagen management trainee. "It was like North Korea without labour camps."[11]

Micromanagement of the company's US offerings by executives in Germany may have caused US sales to lag, some experts said. It had no SUV priced under $30,000 in the US market until 2008 – even though SUVs had been the fastest-growing segment since 2000. Likewise, Volkswagen management had insisted on offering the Phaeton, an expensive sedan priced at $80,000, which flopped.[12] US auto dealers complained that it took years to convince German Volkswagen developers to install cup holders big enough to accommodate takeout coffee in their cars. To the Germans, coffee was drunk at a table. "It was impossible to explain in Wolfsburg that our cup holders didn't work," said Walter Groth, a former Volkswagen employee in the US.

Strategy 2018

By 2007, Winterkorn had announced Volkswagen's plan to become the biggest automaker in the world, which depended heavily on selling diesel cars in the United States. The plan was backed by the forceful personality of VW Chairman Piëch, who was largely responsible for building the company to its current status. Not only did the plan call for VW to nearly double its car sales to 10 million cars per year, it demanded a dramatic improvement in VW's often-lagging profit margins. "In 2018, the Volkswagen Group aims to be the most successful and fascinating automaker in the world," announced the Volkswagen 2009 annual report.[13]

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10 Andreas Cremer and Tom Bergin "Fear and respect: VW's culture under Winterkorn," October 10, 2015, http://www.reuters.com/article/us-volkswagen-emissions-culture-idUSKCN0S40MT20151010

11 Jack Ewing and Graham Bowley, "The Engineering of Volkswagen's Aggressive Ambition," New York Times, December 13, https://www.nytimes.com/2015/12/14/business/the-engineering-of-volkswagens-aggressive-ambition.html

12 Robert Lussier "Management Fundamentals: Concepts, Application, Skill Development," Cengage Learning 2011, pg. 180

13 Volkswagen Annual Report 2009, http://annualreport2009.volkswagenag.com/servicepages/search.php?q=fascinating&pageID=33738&cat=b

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Bob Lutz, a former vice chairman of General Motors, who knew Piëch as a competitor, described his management style as "a reign of terror."[14] He said he once sat next to Piëch at an industry function and the Volkswagen scion explained how he got things done. Piëch told him, when he wanted some ground-breaking design work done, he would call all his engineers into a room and tell them that if they couldn't invent a new way to do the job in six weeks, they would all be fired.

"Piëch said, 'You know your problem, Mr. Lutz? You're way too soft. When I want something, I get it.'" Lutz recounted. He said, "The Volkswagen culture was 'You get it done or it's off with your head.' At Volkswagen all the motivation is fear driven."[15] Winterkorn was also known for a dictatorial tone. One unnamed former VW executive told Reuters:

> *Even in public Winterkorn ordered very senior staff around. A video shot at the Frankfurt motor show four years ago gives a glimpse of the man's style. The video… shows him inspecting a new model from South Korean rival Hyundai, surrounded by a posse of dark-suited managers. He circles the car, inspecting the locking mechanism on its tailgate, and then climbs into the driver's seat. First, he strokes the interior trim, then he adjusts the steering wheel and discovers something that displeases him – it moves silently, unlike on VW or BMW models. Bischoff!" he barks in the footage – no first names or honorifics – summoning VW design chief Klaus Bischoff. "Nothing makes a clonking sound here," he says grumpily, pointing to the wheel.*[16]

Lutz said he believed that Strategy 2018's goal of producing diesel cars that could rack up millions in sales in the United States came down as an order from Piëch – and engineers were told to "Do it or else." Lutz wrote:

> *He just says, "You will sell diesels in the U.S., and you will not fail. Do it, or I'll find somebody who will." The guy was absolutely brutal. I imagine that at some point, the VW engineering team said to Piëch, "We don't know how to pass the emissions test with the hardware we have." The reply, in that culture, most likely was, "You will pass! I demand it! Or I'll find someone who can do it!"*[17]

Setting such ambitious goals – even if they can never be met – is a timeworn management strategy to increase internal drive for achievement. But Yale management professor David Bach said there must be limits set by the companies' top management on how far one can go to achieve those goals. "Stretch goals are very useful," he said. "Precisely because they generate a lot of pressure, you have to make sure they are coupled with a clear sense of what the boundaries are…You should never let the goal itself get the better of you. That was just missing at Volkswagen."[18]

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14 Telephone interview with Bob Lutz April 15, 2016

15 *Ibid*.

16 Andreas Cremer and Tom Bergin "Fear and respect: VW's culture under Winterkorn," October 10, 2015, http://www.reuters.com/article/us-volkswagen-emissions-culture-idUSKCN0S40MT20151010

17 Bob Lutz, "One Man Established the Culture That Led to VW's Emissions Scandal," Road and Track, November 4, 2015, https://www.roadandtrack.com/car-culture/a27197/bob-lutz-vw-diesel-fiasco/

18 Jack Ewing, "Faster, Higher, Farther: The Volkswagen Scandal," 2007, W.W. Norton & Company, pg. 151

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Exhibit 3. Martin Winterkorn inspected a Porsche at a car show in 2008. Source: Marian Murat/European Press photo Agency

The Rise and Fall of "Clean Diesel"

Dirty, Smelly, Nasty

Cars with diesel engines had never caught on in America where they were perceived as belching stinky, sooty exhaust fumes. Diesels were popular in Europe where smog regulations were looser and the government subsidized diesel vehicles through lower taxes. To meet Strategy 2018, Volkswagen vowed to popularize diesels in the US – and so began the "clean diesel" campaign.

Diesel did indeed release fewer greenhouse gases (CO_2) than petrol engines, but it released other noxious gases. Carmakers had become proficient at trapping the particulates that came out of the exhaust pipe as brown smoke, but nitrogen oxides (NOx) gases were harder to control. NOx emissions were composed of several pollutants including nitric acid, nitrous acid and, most harmfully, nitrogen dioxide. High concentrations of nitrogen dioxide could irritate the lungs and provoke respiratory symptoms such as coughing and wheezing. Long-term exposure was linked to the development of asthma in children, heart disease, poor birth outcomes, diabetes, cancer and early mortality.[19] NOx gases were also contributors to the formation of acid rain, ozone and smog.

While BMW had managed to find ways to trap diesel pollutants and meet US standards, it came at a cost – reducing fuel efficiency, and adding to the price of the car. "You have power, you have energy, you have emissions: you get to choose two of them," said Don Hillebrand, director

19 United States Environmental Protection Agency, "Integrated Science Assessment for Oxides of Nitrogen – Health Criteria," E.P.A. website, January 28, 2016, https://cfpub.epa.gov/ncea/isa/recordisplay.cfm?deid=310879

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of energy systems research at Argonne National Laboratory in Illinois, explaining the inherent trade-off between power, fuel-efficiency and low emissions in diesel engines.[20]

As described by Jack Ewing in *Faster, Higher, Farther*, VW set a tight deadline for the release of its new US diesels with a completely redesigned engine known as the EA 189. Yet the problems with the emissions system were not yet worked out. In 2006, designers realized that the new engine's exhaust gas recirculation system would cause the particle filters to wear out prematurely. They didn't want customers having to replace the filters all the time, but neither did they want to add additional emissions equipment. Engineers were not able to solve the dilemma "within the allocated timeframe and budget," VW chairman Hans Dieter Poetsch would later testify. A VW engineer quoted by Ewing put it more bluntly: "It was a bad plan," he said.[21]

Rather than solving the problem, the engineers turned to what became known as a "defeat device" to cheat emissions regulations. This was actually not a device at all but a few lines of software code running inside the engine control unit, alongside millions of other lines of code. The software detected driving conditions (e.g. whether the steering wheel was being turned) to determine whether the car was being emissions-tested in a lab or driven on the road. If it determined the car was not being tested, the unit adjusted operating functions like how often fuel was injected into the engine and how much exhaust passed through the engine's Exhaust Gas Recirculation system – which in turn would influence its NOx emissions. The software also affected variables like fuel consumption, performance and how quickly the NOx filter wore out.[22] The defeat device was designed to turn on maximum emission controls only when the car's computer determined that it was on rollers undergoing emissions tests. When on the road, the controls were turned down and the cars emitted many times the pollution produced in testing situations.

20 Tia Ghose, "Volkswagen Scandal: Why Is It so Hard to Make Clean Diesel Cars," Live Science, September 24, 2015, https://www.livescience.com/52284-volkswagen-scandal-clean-diesel-challenges.html

21 Jack Ewing, "Faster, Higher, Farther: The Volkswagen Scandal," 2007, W.W. Norton & Company, pg. 119

22 Megan Geuss, "A year of digging through code yields "smoking gun" on VW, Fiat diesel cheats," Arstechnica, May 28, 2017, https://arstechnica.com/cars/2017/05/volkswagen-bosch-fiat-diesel-emissions-cheats-cracked-open-in-new-research/

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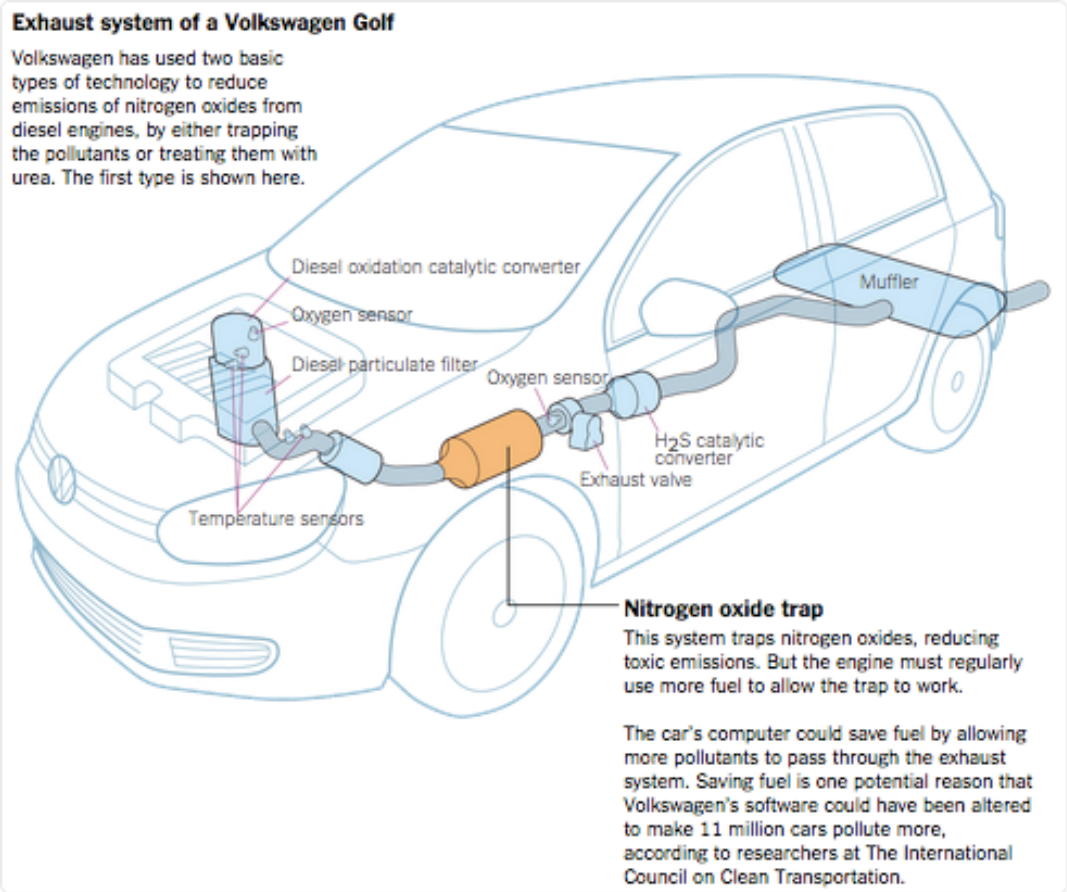

Exhibit 4. How the Volkswagen cheat device worked. Source: New York Times

In late 2008, when Volkswagen introduced its clean diesel campaign in the US, the launch was accompanied by a major advertising campaign designed to repair the reputation of diesel engines. It was an ad blitz that went on for years, with dozens of cute and catchy television commercials. In one, Volkswagen featured three humorous "old wives" dispelling "old wives' tales" about diesel engines being dirty, noisy and smelly. One asks the others how they like her new VW Golf. "I think it's beautiful, but aren't diesels dirty?" the other wife asks. "It used to be dirty, but now it's 2015," says the car-owning wife, holding up her white scarf to the exhaust pipe, which comes away clean. "See how clean it is?" "Okay," says the other wife, "it's not dirty but you still have a dirty mind." A caption flashes on the screen: "Volkswagen TDI Clean Diesel. Like really clean diesel."

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Exhibit 5. "Old wives" in the Volkswagen clean diesel commercials. Source: Volkswagen/Youtube

In Europe, where diesel was cheaper than petrol and relatively high fuel prices made customers efficiency conscious, Volkswagen could use fuel economy as a main selling point for its vehicles. But in the US petrol ("gasoline") was cheap and diesel was slightly more expensive. Volkswagen wanted to use the environmental credentials of its new EA189 engine to compete with Toyota's Prius Hybrid. Clean diesel was touted as the VW's top selling feature.

Humbling the Competition

When Bob Lutz, then head of Global Product Development at General Motors, heard that Volkswagen was releasing a new line of affordable, "clean diesel" cars in 2008, he vowed that his company would match the impressive new technology. So he went to his engineers and asked them to create a similar car that could pass tough US pollution standards, while still delivering high performance and affordability. The team went to work on it but their results mystified him: "We tried and tried and tried. We used the same suppliers. We used the same emissions control devices. We couldn't come anywhere close. I was beating up on my guys and they kept saying: 'It can't be done.'"[23] At the time, Lutz chalked up Volkswagen's accomplishment to the marvels of German engineering.

23 Telephone interview with Bob Lutz April 15, 2016

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Busted

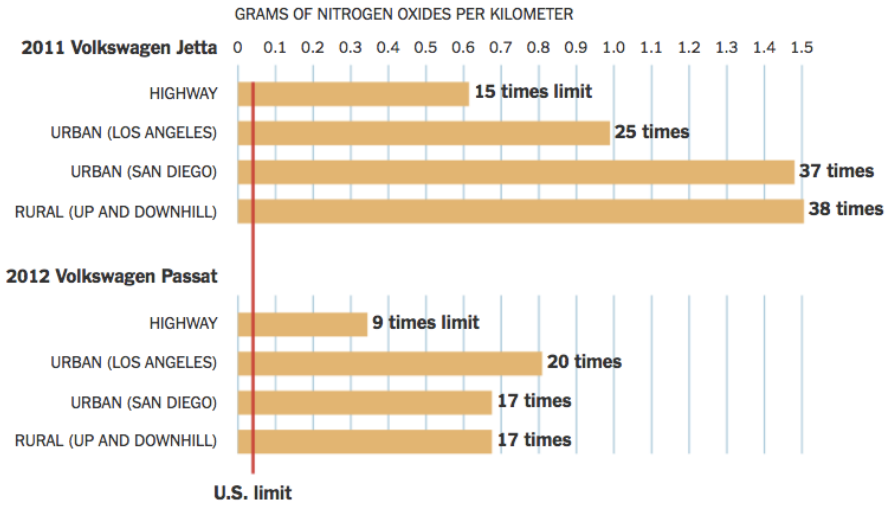

Exhibit 6. Results of tests by West Virginia University researchers. Source: Arvind Thiruvengadam, Center for Alternative Fuels and Emissions at West Virginia University

The new VW models passed the tough US emissions tests with flying colours for seven years. The company's deception might never have been caught if it weren't for three graduate students from West Virginia University who were testing diesel emissions in 2013 using a funky-looking homemade device that hung out the back of a car's tailpipe while it drove on the road. The students, at the West Virginia Center for Alternative Fuels, Engines and Emissions, were testing to see how well diesel emissions equipment performed in a variety of driving scenarios, such as on the freeway and in hilly city conditions. When they checked their data, they were shocked to see that the VWs performed poorly in almost all situations. Their other test car, a BMW SUV, did well in most conditions. But when they tested the cars in the workshop on rollers, all the cars easily met emissions requirements.

Even after these findings, nothing much happened. The students published their 117-page study in 2014 without even naming the brands of the cars tested. "We conducted the study out of pure academic interest and not to implicate any manufacturer. The results of the study simply provided an indicator to a possible problem," said student Arvind Thiruvengadam.[24]

When the California Air Resources Board (CARB), one of the country's toughest pollution regulators, heard about the results, they asked Volkswagen to explain them. They had a good working relationship with Volkswagen and expected some kind of a technical problem that could be easily resolved. "We were finding things that just didn't make sense," said CARB's deputy executive officer Alberto Ayala. "My premise was 'Let's work together to solve this problem."[25]

24 Sruthin Lal, "Indian scientist, W.Va. team helped expose V.W., Hindustan Times, October 3, 2015, http://www.providencejournal.com/article/20151003/ENTERTAINMENTLIFE/151009813

25 Jack Ewing, "Faster, Higher, Farther: The Volkswagen Scandal," 2007, W.W. Norton & Company, pg. 174

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Obfuscation

But when asked to explain, Volkswagen's US managers were evasive. At meetings with California officials to discuss the results of the study, they tried to "discredit the findings," said Stanley Young, CARB communications director. They held one meeting after another without ever seeming to get anywhere: "They were recalcitrant. It was a range of issues. Every time it was something different."[26] VW said that the study's "calibration was off" and that the discrepancies had to do with "conditions under which the test was done," Young said.

After months of negotiations, Volkswagen produced its own test results and said the excess in NOx emissions came from "various technical issues and unexpected in-use conditions," according to the EPA.[27] The company then volunteered to recall vehicles and do a minor fix. Yet the repair didn't correct the problem. CARB demanded the company's data on the results of the fixes, but got no answer. So it did more of its own testing.

Emails later obtained by investigators showed the panic going on inside Volkswagen as it scrambled to come up with a story. Some of these were quoted in court filings (without naming the employees who sent them). In late May 2015, Michael Hennard, senior manager of emissions compliance at VW in the U.S., wrote to his superiors: "Please be aware that this type of action from California ARB staff/management is not a normal process. We are concerned that there may be possible future problems/risks involved."[28] In response he was admonished by a superior for being too frank in an email, according to lawsuits filed since.[29] "We just need a plausible explanation as to why the emissions are still high!!!" one Volkswagen employee wrote in April 2015. On May 12, another email noted, "We need a story for the situation!" One in June cautioned: "We must be sure to prevent the authority from testing Gen 1 (the first generation engine). If Gen 1 goes onto the roller at the CARB, then we'll have nothing more to laugh about!!!!" It wasn't possible to tell whether the emails were from the same or different employees.[30]

A Confession

On September 3, 2015, Volkswagen's US managers changed their tune. A group of managers, including several mid-level managers from its US emissions unit, held a meeting with CARB officials and admitted that VW diesel cars had different settings for when the vehicle was being tested than for normal usage. The software was programmed to turn on maximum emission controls only when the car's computer determined that it was on rollers undergoing emissions tests. When on the road, these were turned down and the cars emitted up to 40 times as much pollution as in testing situations.

The public didn't learn about the problem until the EPA issued a formal notice of violation two weeks later on September 18, 2015: "Put simply, these cars contained software that turns off

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26 Timothy Gardner, Paul Lienert and David Morgan, "After year of stonewalling, Volkswagen stunned U.S. regulators with confession," Reuters, September 24, 2015, http://www.reuters.com/article/us-usa-volkswagen-deception-insight-idUSKCN0RO2IP20150924

27 *Ibid*.

28 Jack Ewing, "Faster, Higher, Farther: The Volkswagen Scandal," 2007, W.W. Norton & Company, pg. 184

29 *Ibid*.

30 Patrick McGee, "Unsealed papers in VW scandal reveal panic among engineers", Financial Times, September 9 2016, https://www.ft.com/content/9bac3826-76af-11e6-bf48-b372cdb1043a

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emissions controls when driving normally and turns them on when the car is undergoing an emissions test," said Cynthia Giles, an enforcement officer at the EPA.[31]

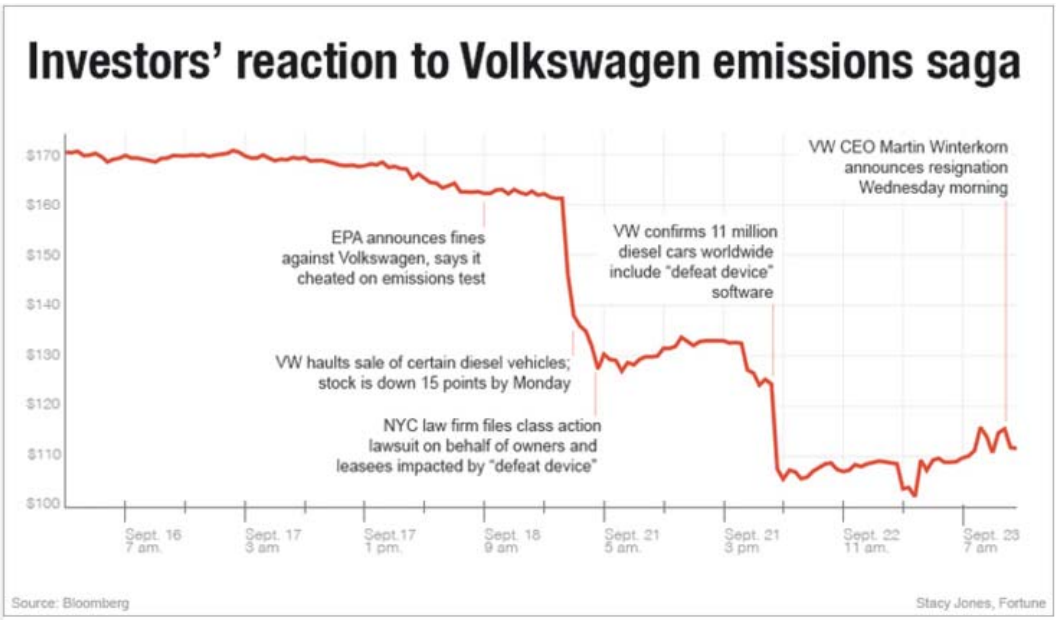

Exhibit 7. Source: Stacy Jones/ Fortune / Bloomberg

The EPA said almost 450,000 diesel cars sold in the US would be recalled. A week later, on Friday September 21st, the problem mushroomed as Volkswagen issued a statement admitting that the devices had been installed on 11 million diesel vehicles sold worldwide over an eight-year period. "Things just went from catastrophic to apocalyptic," opined the auto industry blog, Jalopnik.[32]

By the following Monday, Volkswagen's stock price had lost 20% – from $167 a share to $133. By the end of September it would fall to $104, a combined fall of 37%,[33] wiping $20 billion off its market capitalization.[34]

Lawsuits started pouring in – from regulators at federal, state and international levels, from shareholders and from customers. Criminal prosecutions were launched. Dealers began demanding to know how they would satisfy customers' concerns about their pollution-belching engines. Eco-conscious customers held picket signs at court hearings, demanding that VW buy back their cars. The company was excoriated by the press. Newspapers dubbed the scandal

31 Timothy Gardner, Bernie Woodall, "Volkswagen could face $18 billion penalties from EPA" Reuters, September 18, 2015, http://www.reuters.com/article/us-usa-volkswagen/volkswagen-could-face-18-billion-penalties-from-epa-idUSKCN0RI1VK20150918

32 Máté Petrány, "Volkswagen Admits 'Defeat Device' On 11 Million Engines Worldwide, Sets Aside $7.2B" Jalopnik, September 22, 2015, http://jalopnik.com/volkswagen-admits-it-cheated-with-11-million-engines-s-1732283322

33 Google Finance, Volkswagen AG stock price listings, https://www.google.com/finance/historical?cid=673455&startdate=Sep%203%2C%202015&enddate=Sep%202%2C%202016&num=30&ei=4P2pWcnNCMOo2AburIjgAQ&start=240

34 Investopedia, "VW Scandal: How Has It Impacted Volkswagen's Stock?" October 5, 2016, http://www.investopedia.com/news/vw-scandal-how-has-it-impacted-volkswagens-stock-vlkay/

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"Dieselgate." The *New York Times* editorialized: "What was Volkswagen Thinking?" *The Nation* wrote: "Volkswagen Lied and Cheated 11 Million Times. Will Anyone Go to Jail for That?"

Top Volkswagen officials apologized for the deception. "I personally am deeply sorry that we have broken the trust of our customers and the public," said Winterkorn, who resigned within days of the scandal breaking.[35] Michael Horn, head of Volkswagen's US division, was more blunt: "Our company was dishonest with the EPA and the California Air Quality Board and with all of you," he told an audience in the US. "In my German words, we have totally screwed up."[36]

Repercussions

Knowledge at the Top?

Exhibit 8. Matthias Müller was appointed Volkswagen CEO in the wake of the emissions scandal. Source: Reuters

Top company officials claimed they never knew about the cheat devices, which they said were the work of a few rogue engineers. On September 21st, Winterkorn claimed the "irregularities" in the diesel engines were due to "terrible mistakes made by only a few." He said: "I am not aware of any wrongdoing on my part."[37] Horn likewise denied corporate responsibility: "This was a couple of software engineers who put this in for whatever reason," he told a hearing by a subcommittee of the House of Representatives. He confirmed that VW had suspended three

35 Jack Ewing and Coral Davenport, "Volkswagen to Stop Sales of Diesel Cars Involved in Recall," New York Times, September 20, 2015 https://www.nytimes.com/2015/09/21/business/international/volkswagen-chief-apologizes-for-breach-of-trust-after-recall.html

36 Reuters Video, "Volkswagen: 'We have totally screwed up'" September 22, 2015, https://www.nytimes.com/video/business/international/100000003928968/volkswagen-we-have-totally-screwed-up.html

37 Peter J. Henning, "The Potential Criminal Consequences for Volkswagen," New York Times, September 24, 2015, https://www.nytimes.com/2015/09/25/business/dealbook/thepotential-criminal-consequences-for-volkswagen.html

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employees, but refused to name them, citing German employment law. "To my understanding, this was not a corporate decision. This was something individuals did."[38]

"I cannot accept VW's portrayal of this as something by a couple of rogue software engineers," replied Rep. Chris Collins (R-N.Y.). "Suspending three folks — it goes way, way higher than that."[39] Joan Claybrook, a former administrator of the National Highway Traffic Safety Administration, agreed: "There are not rogue engineers who unilaterally decide to initiate the greatest vehicle emission fraud in history. They don't act unilaterally," she said. "They have teams that put these vehicles together. They have a review process for the design, testing and development of the vehicles."[40]

VW refused to name the suspended employees or even disclose the reasons for the suspensions, but the media confirmed that, by mid-October 2015, about 10 people had been suspended in relation to the investigation.[41] None were top managers.

In late September 2015, VW hired the US law firm Jones Day to conduct an internal investigation into how the deception came about and said it would make the results public. "We're working hard to find out exactly what happened," said Winterkorn in the days after the scandal became public. "To do that, we're putting everything on the table as quickly, rigorously and transparently as possible. We owe that to our customers, our employees and the public."[42] The company appointed Christine Hohmann-Dennhardt, a former Daimler executive, to oversee integrity and legal affairs and ostensibly to straighten out the company's processes. She was the first woman ever to sit on the VW Board.

Company managers vowed to change VW's culture. "This company has to bloody learn and use this opportunity in order to get their act together, and 600,000 people worldwide have to be managed in a different way," Horn told a congressional hearing.[43] Bernd Osterloh, a supervisory board member, wrote in a letter to staff, "We need in future a climate in which problems aren't hidden but can be openly communicated to superiors. …We need a culture in which it's possible and permissible to argue with your superior about the best way to go."[44]

But as months of investigations, legal hearings and negotiations unfolded, the company's actions repeatedly drew questions about whether Volkswagen was serious about changing its ways:

- After CEO Winterkorn resigned, the board chose one of his long-time lieutenants, Porsche CEO Matthias Müller, to replace him. Müller had been head of Volkswagen product

38 Jim Puzzanghera and Jerry Hirsch, "VW exec blames 'a couple of' rogue engineers for emissions scandal," Los Angeles Times, October 8, 2015, http://www.latimes.com/business/autos/la-fi-hy-vw-hearing-20151009-story.html
39 *Ibid*.
40 *Ibid*.
41 Richard Milne, "Volkswagen has suspended 10 senior executives in emissions probe," Financial Times, October 22, 2015, https://www.ft.com/content/337b86de-78c8-11e5-933d-efcdc3c11c89
42 Martin Winterkorn, "Video statement Prof. Dr. Martin Winterkorn," Volkswagen, September 22, 2015 https://www.youtube.com/watch?v=wMPX98_H0ak
43 Andreas Cremer and Tom Bergin "Fear and respect: VW's culture under Winterkorn," October 10, 2015, http://www.reuters.com/article/us-volkswagen-emissions-culture-idUSKCN0S40MT20151010
44 *Ibid*.

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planning from 2007 to 2010, the same time Volkswagen's "clean diesel" campaign was being launched.[45]

- After promising transparency, the company refused to turn over internal emails to state attorneys general in the U.S., citing German privacy laws.[46]
- The company refused to release the results of the internal investigation commissioned from the law firm Jones Day. Despite its earlier promises that the report would be made public, chairman Hans Dieter Poetsch said releasing the report would result in massive fines for the company. "…for legal reasons Volkswagen is prevented from publishing any such report," he said.[47]
- Müller appeared to remain in denial about the gravity of the company's actions when he responded to a National Public Radio reporter's question: "We didn't lie. We didn't understand the question first. And then we worked since 2014 to solve the problem." After prompting international outrage, Muller said he had misspoken and asked to do the interview again.[48]
- Christine Hohmann-Dennhardt, hired to handle integrity and legal affairs, lasted little more than a year. Volkswagen issued a statement that said she had toughened the company's compliance structures but was leaving "due to differences in their understanding of responsibilities and future operating structures within the function she leads."[49] She left with a €10 million payout.[50]

Investigators trying to get to the bottom of who knew about the cheat devices inside the company got little help from Volkswagen. "Our patience with Volkswagen is wearing thin," said New York's attorney general Eric T. Schneiderman in January 2016. "Volkswagen's cooperation with the states' investigation has been spotty — and frankly, more of the kind one expects from a company in denial than one seeking to leave behind a culture of admitted deception."[51]

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45 Danny Hakim and Jack Ewing, "Matthias Müller in the driver's seat at Volkswagen," New York Times, October 1, 2015, https://www.nytimes.com/2015/10/02/business/international/matthias-muller-in-the-drivers-seat-at-volkswagen.html

46 Danny Hakim and Jack Ewing, "VW refuses to give states documents in emissions hearings," New York Times, January 8, 2016, https://www.nytimes.com/2016/01/09/business/vw-refuses-to-give-us-states-documents-in-emissions-inquiries.html?_r=0

47 Jay Ramey, "VW still won't publish diesel investigation findings, citing more fines," Autoweek, May 15, 2017, http://autoweek.com/article/vw-diesel-scandal/vw-still-wont-publish-diesel-investigation-findings-citing-more-fines#ixzz4rf5rgHBe

48 Sonari Glinton, 'We Didn't Lie,' Volkswagen CEO Says Of Emissions Scandal," January 11, 2016 NPR, http://www.npr.org/sections/thetwo-way/2016/01/11/462682378/we-didnt-lie-volkswagen-ceo-says-of-emissions-scandal

49 Prashant S. Rao, "VW Executive, Hired to Help Overhaul Carmaker's Culture, Is to Leave," New York Times, January 26, 2017, https://www.nytimes.com/2017/01/26/business/volkswagen-germany-compliance-hohmann-dennhardt-diesel.html?_r=0

50 Graham Ruddick, "VW risks new pay row as compliance chief lands €10m payoff," The Guardian, January 31, 2017, https://www.theguardian.com/business/2017/jan/31/volkswagen-head-of-legal-affairs-christine-hohmann-dennhardt-10m-pay-off

51 Danny Hakim and Jack Ewing, "VW refuses to give states documents in emissions hearings," New York Times, January 8, 2016, https://www.nytimes.com/2016/01/09/business/vw-refuses-to-give-us-states-documents-in-emissions-inquiries.html?_r=0

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Punishment

The Justice Department described Volkswagen's deception as "one of the largest corporate fraud schemes in the history of the United States."[52] The number of lawsuits and charges from regulators to be settled seemed unending. By September 2017, VW had been forced to put aside about $30 billion to cover the costs of the scandal.[53] Agreed upon fines and paybacks for the US alone were estimated by the Associated Press at more than $20 billion.[54] These included about $20,000 per car to buy back or repair the vehicles and compensate their owners for the trouble.[55] On top of that $20 billion, in September 2017, VW Volkswagen put aside another $2.9 billion because the diesel repairs were costing more than expected.[56]

VW agreed to pay $15 billion in civil fines and compensation to owners of cars with 2-litre diesel engines in October 2016, and another $1.2 billion to owners of 3-litre engines in May 2017. The company pleaded guilty to a separate set of criminal charges including fraud and obstruction of justice. This resulted in another $4.3 billion in penalties imposed in a settlement in April 2017.

Eight VW managers faced criminal charges in the US but none were senior managers. Since Germany did not permit extradition of its citizens on criminal charges, most of the managers could not be put on trial. Volkswagen managers in Germany were advised not to travel to the United States lest they be nabbed for prosecution. In January 2017, Oliver Schmidt was arrested as he passed through the US on his way home from a trip to Cuba.[57] Schmidt, who headed VW's US environment and engineering office, appeared at one court hearing wearing shackles and an orange prison jumpsuit. In August 2017, he pleaded guilty to conspiring to mislead regulators, and was expected to be sentenced to up to seven years in prison in January 2018. Schmidt, one of the managers who met with regulators after the discrepancies in NOx emissions were discovered, had deceived them "by offering reasons for the discrepancy other than the fact that VW was intentionally cheating on US emissions tests," said his arrest affidavit.[58]

Another VW employee in the US, James Robert Liang, a German citizen living in California, was sentenced to 40 months in prison in August 2017. Described by prosecutors as a "mild-mannered" diesel engineering expert and family man, he had spent his entire 34-year career at

52 David Shepardson, "Volkswagen set to plead guilty on Friday in emissions case," Reuters, March 9, 2017, http://www.reuters.com/article/us-volkswagen-emissions-idUSKBN16G2ZR

53 Associated Press, "Where things stand in the Volkswagen Emissions scandal," Associated Press, March 11, 2017, https://www.usnews.com/news/best-states/california/articles/2017-03-10/where-things-stand-in-the-volkswagen-emissions-scandal

54 Associated Press, "Where things stand in the Volkswagen Emissions scandal," Associated Press, March 11, 2017, https://www.usnews.com/news/best-states/california/articles/2017-03-10/where-things-stand-in-the-volkswagen-emissions-scandal

55 Jack Ewing, "In the U.S., VW Owners Get Cash. In Europe, They Get Plastic Tubes," https://www.nytimes.com/2016/08/16/business/international/vw-volkswagen-europe-us-lawsuit-settlement.html?module=Promotron&region=Body&action=click&pgtype=article

56 Associated Press, "VW to take $2.9 billion charge over diesel retrofit costs," Associated Press, September 29, 2017, https://wtop.com/europe/2017/09/vw-to-take-2-9-billion-charge-over-diesel-retrofit-costs/

57 Jörn Poltz and Andreas Cremer, "Senior VW managers warned not to travel to U.S.: sources," Reuters, January 13, 2017, http://www.reuters.com/article/us-volkswagen-emissions-extradition-idUSKBN14X175

58 By Adam Goldman, Hiroko Tabuchi and Jack Ewing, "F.B.I. Arrests Volkswagen Executive on Conspiracy Charge in Emissions Scandal," New York Times, January 9, 2017, https://www.nytimes.com/2017/01/09/business/volkswagen-diesel-emissions-investigation-settlement.html

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VW. Prosecutors said he was the one person who knew the fraudulent design inside and out, and had been involved since it was first being designed for VWs in 2006. Liang worked on the design of the clean diesel in Germany then moved to work at a VW testing centre in California, where he shepherded the cars through the testing process.[59]

Liang, 63, made $350,000 a year and lived in a large house in an exclusive neighbourhood of Southern California for which VW paid the rent. His attorney, Daniel Nixon, told the judge "He wasn't the mastermind. He was blindly – blindly – executing misguided loyalty to his employer."[60] Liang agreed to work with prosecutors to help them make the case against others in the company.

In July 2017, US prosecutors inched closer to implicating members of VW's top management. At the behest of the US authorities, German authorities arrested Zaccheo Giovanni Pamio, former head of thermodynamics at Volkswagen's Audi division in Germany. Since Pamio, 60, was an Italian citizen, the German authorities did not afford him the same protection as their own, and allowed him to be extradited. While jailed in Germany, Pamio told authorities that top managers at Audi knew about the emissions cheats as early as 2006, according to his lawyer Walter Lechner,[61] but Lechner did not name the managers Pamio referred to.

Back then, Winterkorn had been chief of Audi in 2006, with Matthias Müller as Audi's head of project management. "The net is drawing closer on the present and former management board," said Christian Strenger, former chairman of the International Corporate Governance Network, an advocacy group. "The evidence is revealing more and more they were part of the whole affair more or less from the beginning."[62]

These suspicions matched charges brought against Volkswagen top managers made by several US state attorneys general, who settled numerous lawsuits against Volkswagen in March 2017. The suits claimed that knowledge of the cheating was widespread within the company; that Müller was aware of the emissions issues with the diesel cars as early as 2006, and that Winterkorn knew about them too.[63] "The allegations against Volkswagen, Audi and Porsche reveal a culture of deeply rooted corporate arrogance, combined with a conscious disregard for the rule of law or the protection of public health and the environment," New York Attorney General, Eric Schneiderman, said.[64] New York received about $32 million in the settlement.

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59 Eric D. Lawrence, "VW engineer gets 40 months in prison for role in diesel scandal," USAToday, August 25, 2017, https://www.usatoday.com/story/money/cars/2017/08/25/vw-engineer-gets-40-months-prison-role-diesel-scandal/602584001/

60 Robert Snell, "VW engineer's 40-month sentence 'sends ... message,'" The Detroit News, August 25, 2017, http://www.detroitnews.com/story/business/autos/foreign/2017/08/25/vw-engineer-laing-sentencing-diesel-emissions-scandal/104935982/

61 Jack Ewing, "Audi Engineer Implicates Superiors in Diesel Case, Lawyer Says," New York Times, August 21, 2017, https://www.nytimes.com/2017/08/21/business/audi-volkswagen-diesel-emissions.html

62 *Ibid*.

63 Associated Press, "Volkswagen CEO knew of emissions trouble 10 years ago, prosecutor says," Los Angeles Times, July 19 2016, http://www.latimes.com/business/autos/la-fi-hy-vw-audi-porsche-20160719-snap-story.html

64 Press release from New York Attorney General, July 29, 2016, https://ag.ny.gov/press-release/ny-ag-schneiderman-massachusetts-ag-healey-maryland-ag-frosh-announce-suits-against

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Meanwhile, prosecutors in Germany expanded their probe from 21 to 37 suspects in January of 2017, saying they had evidence Winterkorn may have known of the cheating earlier than he had claimed.[65] They added Müller to their list of suspects and raided his office for documents.[66]

The quest to show that top executives had knowledge of the emissions problems fed another set of lawsuits filed by investors who claimed that they lost money because Volkswagen managers failed to warn them of impending problems in a timely manner. In September 2016, shareholders, hoping to meet a one-year cut-off for filings, flooded a German regional court with more than 1,400 lawsuits demanding €8.2 billion in damages for lost investments. Other shareholders filed a class action suit in the United States. In particular, investors wanted to know when top Volkswagen executives knew about the defeat devices and the investigations launched by regulators. In March 2016, Volkswagen acknowledged that CEO Winterkorn had received memos about the US investigation of the emissions discrepancies as early as May 2014.[67] "Whether and to which extent Mr. Winterkorn took notice of this memo at that time is not documented," the company said in a statement.[68]

Investigators also questioned whether top managers were involved in approving the design of the devices, given the fact that Volkswagen was known as a top-down company where executives got involved in the minutiae of engineering decisions. Documents uncovered by the *New York Times* in May 2017 suggested that Winterkorn and Müller, who was then head of product planning, had attended a meeting where the emissions problem was the main agenda item back in 2007, when the engines were still being designed. The *Times* obtained two different email attachments containing versions of the presentation that was supposed to have been given to managers at a meeting on November 8, 2007.[69] Both versions acknowledged that the new engine could not meet US standards for NOx emissions and suggested fixes that would cost about €270 per car. One included an appendix that described the specifics of the defeat device. This version, which was sent to managers just below the level of the management board, contained a warning not to forward the email to anyone else.[70] What happened at the meeting was unclear. Volkswagen said there was no evidence that either top manager saw the presentation. "The Volkswagen Group is aware of the documents," the company responded in a statement, "and they do not support the inference that Matthias Müller knew about efforts to develop and use the defeat device."[71]

Other documents seen by the *New York Times* indicated that engineers had sought top management approval to implement the defeat devices in 2007.[72] One status report in May 2007

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65 Associated Press, "German Prosecutors Widen Volkswagen Emissions Probe," VOA, January 27, 2017, https://www.voanews.com/a/german-prosectuors-widen-volkswagen-emissions-probe/3695403.html

66 Jack Ewing, "VW Engineers Wanted O.K. From the Top for Emissions Fraud, Documents Show," New York Times, May 17, 2017, https://www.nytimes.com/2017/05/17/business/volkswagen-muller-diesel-emissions.html

67 Megan Geuss, "Volkswagen details what top management knew leading up to emissions revelations," Cars Technica, March 3, 2016, https://arstechnica.com/cars/2016/03/volkswagen-says-ceo-was-in-fact-briefed-about-emissions-issues-in-2014/

68 *Ibid*.

69 Jack Ewing, "VW Engineers Wanted O.K. From the Top for Emissions Fraud, Documents Show," New York Times, May 17, 2017, https://www.nytimes.com/2017/05/17/business/volkswagen-muller-diesel-emissions.html

71 *Ibid*.

72 *Ibid*.

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noted that approval from top managers was needed for "off-cycle functionality" – another way of saying a car would function differently on the road than at a testing station. There were no indications as to whether approval was given.

VW indicated in March 2016 that its managers did not think the defeat device would be that big a deal – that they did not pay that much attention to the issue, based on advice that past penalties for companies with emissions violations had been around $100 million dollars. "Volkswagen was advised that in the past [US emissions penalties] were not especially high for a company the size of Volkswagen," the company said in a statement explaining why top managers were not involved in the issue. [73]

Another issue that continued to hang over Volkswagen was what to do about emissions emitted by the 9 million diesel cars that it had sold in Europe. Unlike the US, where vehicle owners were given compensation, European car owners did not have the same protection. Volkswagen claimed the technology it installed was not a violation of EU law, which was less stringent about emissions and had many loopholes. Nonetheless, the company offered free "fixes" to improve the emissions of vehicles already on the road in Europe. However, many owners claimed that the "fixes" caused poor vehicle performance and lower gas mileage.[74] A group of 41,000 Volkswagen owners joined a lawsuit in Britain seeking damages for the problems caused by Volkswagen's emission technology and the fix. Volkswagen replied that it had done fixes in over 5 million vehicles throughout Europe without significant complaints. "There is no systemic problem," the company said in a statement. "The overwhelming majority of our customers have been fully satisfied."[75]

Volkswagen was not the first car company to be prosecuted over cheat devices. In 1998, both Ford and Honda paid several hundred million dollars in fines and remediation for installing devices that disabled the emissions equipment in their cars. Regulators alleged that Honda disabled a misfire monitor that would normally have turned on an engine light when emissions controls were malfunctioning. Ford allegedly installed a defeat device in vans that improved fuel consumption but increased emissions. In these cases, the companies cooperated with the EPA and settled out of court.

Back in 1974, Volkswagen itself had been fined $120,000 to settle a complaint by the EPA that it failed to disclose devices that modified emissions controls on petrol engines.[76] What was different about the 2015 case was the emphasis on selling its products as "clean diesel" and the lengths its engineers went to deceive regulators. "Compared with other run-ins between the EPA and automakers, VW's alleged violation stands out in its brazenness," wrote *Automotive News*.[77]

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73 Nathan Bomey, "Volkswagen: We thought scandal wouldn't cost much," USA Today, March 2, 2016, https://www.usatoday.com/story/money/cars/2016/03/02/volkswagen-emissions-scandal/81214726/

74 Miles Brignall, "Thousands of drivers suffer loss of power following VW emissions 'fix,'" The Guardian, July 12, 2017, https://www.theguardian.com/money/2017/jul/12/drivers-loss-of-power-vw-emissions-fix-class-action

75 *Ibid*.

76 Ryan Beene, "VW emissions defeat device isn't the first," Auto Week, September 24, 2015, http://autoweek.com/article/car-news/vw-emissions-defeat-device-isnt-first

77 Ryan Beene, "VW faced ultimatum from EPA," Automotive News, September 20, 2015, http://www.autonews.com/article/20150920/OEM11/309219947/vw-faced-ultimatum-from-epa11

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In the years before Volkswagen's deception was uncovered, other carmakers were facing enquiries into fatal flaws faults in their vehicles, yet the fines and penalties they ended up paying were far smaller. After it was determined that faulty ignition switches caused more than 100 deaths, GM ended up paying fines and settlements of $2 billion dollars[78] (even though executives knew about the problem and failed to fix it for years). Japanese parts maker Takata plead guilty to charges linked to its manufacture of faulty airbags, which were blamed for 11 deaths and 180 injuries in the US. It settled for $1 billion.[79]

"Premeditation. That's the difference between Volkswagen's faked diesel emission tests and earlier auto scandals. It's what makes this worse than GM's ignition switches, Toyota's runaway cars or Ford-Firestone's exploding tires," wrote Mark Phelan, an auto critic for the *Detroit Free Press*.[80] "Volkswagen set out to cheat emissions tests and sell cars that would damage human health and the environment. The other automakers seemed legitimately baffled and eager to address their crises."

Conclusion: The End of Diesel?

While VW's excess diesel emissions couldn't be directly linked to any one death, studies projected a high death toll nonetheless. A team of researchers led by an aeronautics professor from MIT estimated that, in Europe, the excess pollution released by VW diesels skirting the pollution limits would eventually result in 1,200 premature deaths.[81] In the US, where diesel vehicles were far less common, VW emissions excesses were expected to cause 60 premature deaths by 2016, according to another study by researchers from Harvard and MIT.[82] In total, the European Environmental Agency estimated that NOx pollution, much of which came from diesel vehicles, caused 75,000 premature deaths in Europe each year.[83]

Diesel exhaust was blamed for major smog crises in European cities, including a spell in 2015 that briefly made Paris the city most polluted city on earth (overtaking Beijing). Cities such as Paris, Madrid, Athens and Mexico City agreed to outlaw diesel vehicles from the city centre by 2025.[84] Britain and France vowed to stop allowing new diesels by 2040. Many German cities faced possible court orders to ban diesel vehicles after an environmental lawsuit.

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78 Jack Ewing and Neal E. Boudette, As VW Pleads Guilty in U.S. Over Diesel Scandal, Trouble Looms in Europe," New York Times, March 10, 2017, https://www.nytimes.com/2017/03/10/business/volkswagen-europe-diesel-car-owners.html

79 *Ibid*.

80 Mark Phelan, "Lies make VW scandal worse than GM, Toyota," Detroit Free Press, September 26, 2015, http://www.freep.com/story/money/cars/mark-phelan/2015/09/26/vw-worse-than-gm-diesel-volkswagen-general-motors/72764466/

81 Guillaume P. Chossière et al., "Public health impacts of excess NOx emissions from Volkswagen diesel passenger vehicles in Germany," Environmental Research Letters, March 3, 2017, http://iopscience.iop.org/article/10.1088/1748-9326/aa5987

82 Steven R H Barrett, et al. "Impact of the Volkswagen emissions control defeat device on US public health," Environmental Research Letters, October 29 2015, http://iopscience.iop.org/article/10.1088/1748-9326/10/11/114005/meta

83 Geoffrey Smith, "An Alarming Number of People Have Died from Dirty Diesel Engines," Fortune, November 30, 2014, http://fortune.com/2015/11/30/diesel-emissions-deaths-europe-ee/

84 *Ibid*.

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As automobile buyers began to recognize the effects on the environment, diesel car sales in Europe plummeted. During the first six months of 2017, sales declined 10% in Britain, 9% in Germany and 7% in France.[85] Many questioned whether Europe's love affair with diesel was over.

Once the problems with Volkswagen diesels came to light, other companies were also suspected of being in violation of emission regulations. Daimler began voluntarily upgrading the emissions controls on its Mercedes diesel cars in summer 2017, in what many saw as an attempt to avoid the kind of penalties VW was facing. In the past, its emissions equipment only operated at its maximum efficiency when the weather was within a narrow temperature range. The upgrade promised to broaden that temperature range.[86]

However, it seemed that while other manufacturers' vehicles might have exceeded emission limits during the driving cycle (emissions vary under different driving conditions, such as going uphill or on a flat road), they had not apparently done as VW and programmed the software to detect when the vehicle was under test conditions and adjusted its emissions in that mode to meet the regulations.

Meanwhile, VW showed signs of resilience. At the beginning of 2017, due to an increase in sales of mostly non-diesel cars in China, Volkswagen claimed the record for having sold more cars than any other automaker in 2016. For that year, its goal of world domination was actually achieved.

85 Jack Ewing, "As Emissions Scandal Widens, Diesel's Future Looks Shaky in Europe," New York Times, July 25, 2017, https://www.nytimes.com/2017/07/25/business/diesel-emissions-volkswagen-bmw-mercedes.html?_r=0

82 Jack Ewing and Prashant s. Rao, "Daimler to Modify 3 Million Mercedes Cars Over Diesel Concerns," https://www.nytimes.com/2017/07/18/business/daimler-diesel-emissions.html

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Appendix: Timeline of the Volkswagen Emissions Scandal[87]

1999 – The first devices designed to evade pollution standards within the VW companies were allegedly developed by engineers for Volkswagen's Audi division as part of an effort to reduce the rattling sound made by diesel engines. To do this, they injected extra fuel into the engine, but this caused NOX emissions to increase, so they designed a device that would cause the car to perform differently under testing conditions.

2005 – Volkswagen wanted to ramp up its sale of diesels in the US, but engineers were unable to find a technical solution that would allow them to comply with pollution regulations within "timeframe and budget." They decided to deploy the cheat devices (already in some Audi engines) to work around the problem by deceiving regulators.

Late 2008 – Volkswagen begins a "clean diesel" marketing push in the United States.

2013 – A team from West Virginia University runs a road test which reveals that two Volkswagens emit far more nitrogen oxide than is legally permitted.

Spring 2014 – The California Air Resources Board (CARB) opens an inquiry into the excess VW emissions. Executives and engineers try to cover up the illegal software by feeding the regulators false and misleading data.

Spring 2015 – VW recalls diesel vehicles in the U.S., telling regulators that a software update will fix the problem. But the fixes don't work; CARB intensifies its inquiry.

September 3, 2015 – Volkswagen confesses to the Environmental Protection Agency (EPA) and CARB that its diesel cars are equipped with defeat-device software that detects when emissions tests are taking place and cranks up pollution controls so that the cars will pass.

September 18, 2015 – The public learns of the fraud after the EPA issues a formal notice of violation, saying it affects almost 500,000 cars in the U.S.

September 21, 2015 – Volkswagen admits devices were installed in 11 million vehicles worldwide. CEO Martin Winterkorn blames the problems on "the terrible mistakes of a few."

September 23, 2015 – Winterkorn resigns, says he is not aware of any wrongdoing on his part

June 2016 – VW agrees to a $14.7 billion settlement in the US to deal with civil complaints and consumer lawsuits. The settlement includes compensation to owners and measures to atone for the cheating, such as installing networks of electric car charging stations.

January 2017 – Volkswagen pleads guilty to conspiring to defraud the United States government and violate the Clean Air Act. The company will pay a $4.3 billion fine. US prosecutors separately indict six former Volkswagen executives, including an engineer arrested at Miami International Airport.

March 2017 – German prosecutors raid offices of Audi. German prosecutors state that 37 people are suspected of taking part in the fraud, including Winterkorn and Müller.

September 2017 – VW forced to put aside $30 billion to cover the costs of the scandal.

87 Adapted from: Jack Ewing, "Engineering a Deception: What Led to Volkswagen's Diesel Scandal," New York Times, March 16, 2017, https://www.nytimes.com/interactive/2017/business/volkswagen-diesel-emissions-timeline.html

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